

Real Time Helmet Violation Detection Using Deep Learning

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BATCH NO: 30



Agenda

- 1. ABSTRACT**
- 2. DOMAIN**
- 3. EXISTING SYSTEM**
- 4. EXISTING ARCHITECTURE**
- 5. PROBLEM STATEMENT**
- 6. LITERATURE SURVEY**
- 7. LITERATURE SURVEY TABLE**
- 8. PROPOSED SOLUTION**
- 9. PROPOSED SYSTEM**
- 10. PROPOSED SYSTEM ARCHITECTURE**
- 11. HARDWARE AND SOFTWARE REQUIREMENTS**
- 12. ENVIRONMENTAL SETUP SCREENSHORT**
- 13. IMPLEMENTATION DETAILS**
- 14. RESULT AND DISSCUSION**
- 15. CONCLUSION AND FUTURE ENHANCEMENT**
- 16. REFERENCE**

ABSTRACT

Artificial Intelligence (AI) is increasingly being applied to solve everyday societal challenges by enabling machines to perform tasks that require human-like intelligence. Within this domain, deep learning has emerged as a powerful technique in computer vision, allowing systems to automatically recognize, classify, and analyze visual data for improved decision-making.

In the existing system, researchers proposed an artificial intelligence multitier framework with a lightweight classifier for helmetless biker detection. Their method follows a two-stage process: first detecting motorbike riders from traffic footage, and then classifying whether they are wearing helmets. By designing a lightweight model, the system reduces computational cost, enabling faster detection with acceptable accuracy. The authors also built custom datasets to train and evaluate the system under different conditions. While this approach achieved promising results, it primarily addressed only helmet detection and was tested under limited scenarios.

ABSTRACT continuation...

The disadvantage of the existing work is that it does not consider broader violations such as triple riding, mobile phone usage, or varying environmental conditions like night and rain. Our proposed work addresses these gaps by utilizing detectnet and YOLO v8 algorithm, while also incorporating robust datasets to cover diverse scenarios. Additionally, the system will generate automated alerts and send them to authorized institutions or organizations, triggering warning messages to riders. This ensures higher accuracy, real-time enforcement, and practical deployment for improving road safety.

Keywords: Artificial Intelligence, Deep Learning, Helmet Detection, Traffic Safety, Computer Vision.

DOMAIN:

ARTIFICIAL INTELLIGENCE (AI) / COMPUTER VISION / DEEP LEARNING

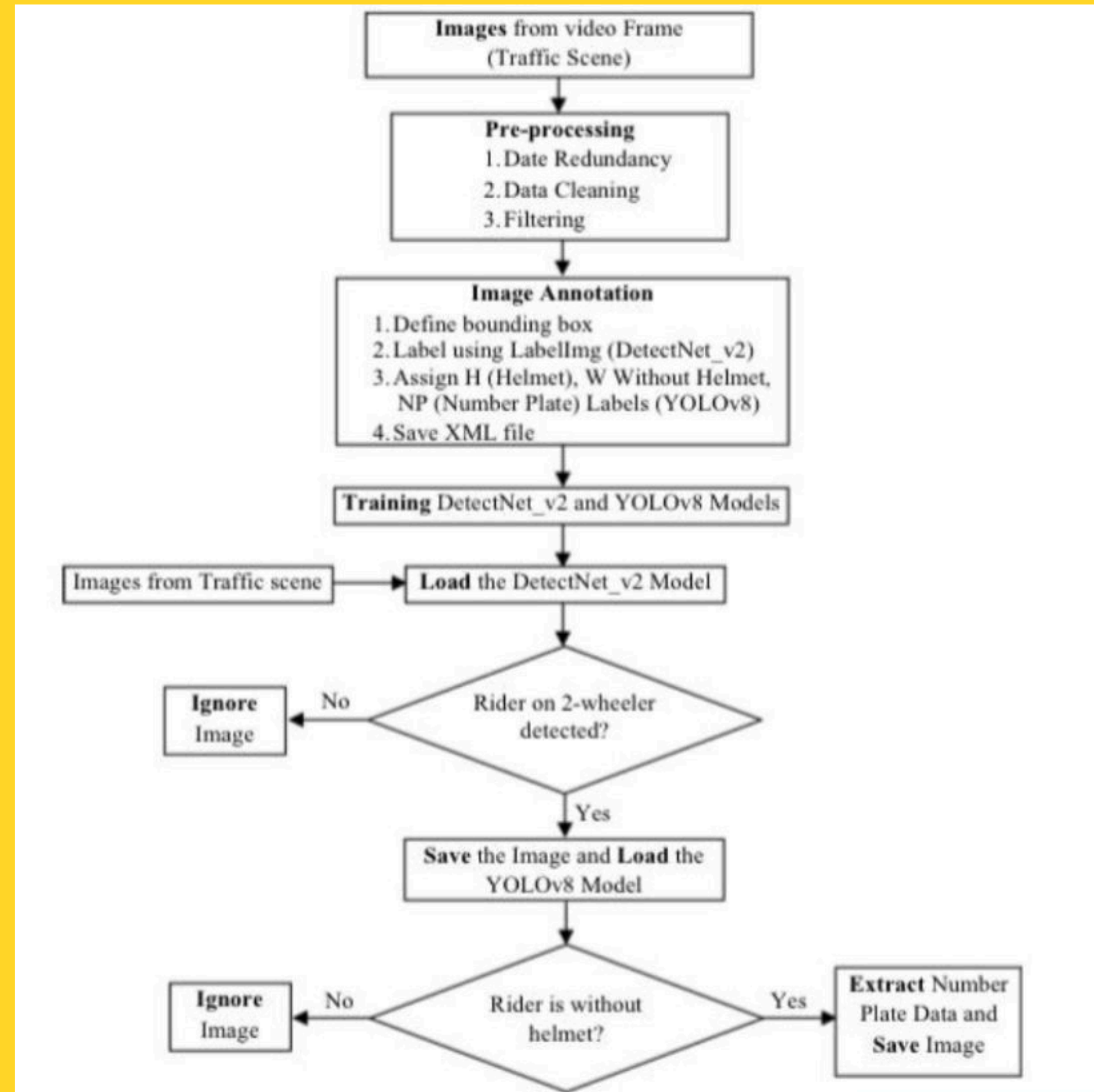
This project falls under the domain of **Artificial Intelligence**, specifically focusing on Computer Vision and Deep Learning. It uses AI techniques to automatically analyze video footage, detect motorcycles, and identify helmet usage in real time, making it a practical application of intelligent image and video analysis for traffic monitoring and road safety enforcement.



Existing System

- In the existing system, helmet violation detection is mostly done manually by traffic police or through basic image processing techniques. **Traffic officers monitor CCTV footage** or roadside cameras and identify violations visually, which is time-consuming, less accurate, and prone to human error.
- Some traditional systems use simple color or shape-based image processing, but they fail under poor lighting, different helmet styles, or complex backgrounds.
- Overall, **existing systems Lack automation**, real-time processing, and accuracy, highlighting the need for an advanced deep learning-based approach to detect helmet violations more efficiently.

EXISTING ARCHITECTURE:



PROBLEM STATEMENT:

- Current AI-based systems use **lightweight** classifiers in a two-stage process — first detecting motorbike riders, then classifying helmet usage.
- These methods focus only on **helmet detection**, often under limited conditions.
- Existing models struggle under challenging conditions like:
 1. **Night-time**
 2. **Rain or poor lighting**
- They also lack automated enforcement features, meaning there is no instant alert to authorities or warning to offenders.

LITERATURE SURVEY:

TITLE : Automated detection of bicycle helmets using deep learning

AUTHORS: Siebert F.W., Riis C., Janstrup K.H., Lin H., et al.

YEAR: 2024

DESCRIPTION:

This study presents a computer vision approach to automatically detect bicycle helmets from roadside camera footage in Copenhagen. A dataset of ~4,000 cyclists, supplemented with web images, was annotated for active cyclist detection and helmet use. The authors trained and tested deep learning models to estimate helmet-wearing rates, comparing results against manual human observations. The system demonstrated near-human accuracy when tested on the same site, with a minimal underestimation of 0.52%. However, accuracy dropped when applying models trained at one site to another, highlighting the challenges of cross-site generalization in real-world deployments.

CONCLUSION:

Automated helmet-use detection is feasible and highly accurate under controlled conditions. However, cross-site transfer remains a limitation, and improving robustness across diverse environments is crucial for large-scale deployment.

TITLE: Effect of helmet use on maxillofacial injuries: a systematic review and meta-analysis

AUTHORS: Stassen H.S., Atalik T., Haagsma J.A., Wolvius E.B., et al.

YEAR: 2024

DESCRIPTION:

This paper systematically reviews and meta-analyzes studies examining the relationship between helmet use and maxillofacial injuries in bicycle and scooter crashes. Fourteen studies were included, of which eleven contributed to the quantitative synthesis. The pooled odds ratio ($OR \approx 0.68$) indicates that helmets significantly reduce the risk of maxillofacial trauma. The included studies were predominantly observational, with variable quality ratings. The review emphasizes that while evidence supports protective effects, research is limited by heterogeneity, older datasets, and lack of focus on e-bikes and e-scooters, which have unique injury patterns and growing urban prevalence.

CONCLUSION:

Helmets provide significant protection against maxillofacial injuries. However, higher-quality and more recent studies, particularly addressing e-bikes and e-scooters, are needed to strengthen evidence and guide policy.

TITLE: AI-Based Helmet Violation Detection for Traffic Management System

AUTHORS: Said Y., Alassaf Y., Ghodhbani R., Alsariera Y.A., et al.

YEAR: 2024

DESCRIPTION:

This work introduces an AI-driven helmet violation detection framework aimed at improving road safety and assisting traffic enforcement authorities. The system is built on a modified PerspectiveNet model with EfficientNet v2 as the backbone, balancing detection accuracy and computational efficiency for real-time surveillance. It is evaluated using public traffic datasets and demonstrates a 95.2% detection accuracy, with reduced model complexity compared to conventional deep learning approaches. The authors highlight scalability in dense urban traffic, capability for real-time operation, and potential for integration into broader intelligent transportation systems.

CONCLUSION:

The study validates the feasibility of using lightweight deep learning models for helmet violation detection in real-world traffic conditions. It emphasizes scalability and efficiency, making the system suitable for real-time deployment in smart cities.

TITLE: DetectNet_v2 + YOLOv8 for helmet violation detection and license plate extraction

AUTHORS: Deshpande U.U., Goh G.K.O.M., Araujo S.D.C.S., et al.

YEAR: 2025

DESCRIPTION:

This paper proposes a two-stage helmet violation detection pipeline tailored for Indian traffic conditions. In the first stage, DetectNet_v2 with a ResNet18 backbone detects motorcycle riders, while the second stage employs YOLOv8 to classify helmet use and detect license plates. An OCR module extracts license numbers for enforcement. A custom dataset of ~1,715 images captured in varied lighting and traffic conditions is used for training and evaluation. The system achieves impressive accuracy: helmet detection (98.56%) and license plate recognition (97.6%), with competitive inference times for near real-time application in urban surveillance systems.

CONCLUSION:

The two-stage pipeline demonstrates robust performance for helmet violation detection and license plate recognition. Its strong accuracy and practical orientation highlight its suitability for deployment in Indian smart-city surveillance networks.

TITLE: Smart Road Surveillance: Helmet Detection and Triple Ride Detection Using CNN and YOLO

AUTHORS: Ranganayaki J., Poojitha V., Dikshith Y.

YEAR: 2025

DESCRIPTION:

This engineering-oriented paper presents a smart road surveillance system combining YOLO and CNN for detecting helmet violations and triple-riding cases on motorcycles. The system includes license plate recognition using Tesseract OCR, enabling automatic identification of violators. The implementation is tailored for resource-constrained environments, focusing on practical deployment in developing regions. The authors describe the system architecture, dataset preparation, and provide screenshots of real-world tests. While performance metrics are not as comprehensive as in research-focused works, the emphasis is on integrating multiple violation detection features into a functional prototype system.

CONCLUSION:

The proposed system demonstrates the integration of helmet and triple-ride detection into a single surveillance framework. Its strength lies in practical feasibility and low-cost deployment, though further validation with larger datasets is needed.

LITERATURE SURVEY TABLE:

TITLE	TECHNIQUE USED	CONCLUSION
AI-Based Helmet Violation Detection for Traffic Management System	Optimized PerspectiveNet with EfficientNet v2 backbone, deep learning-based real-time helmet violation detection.	We conclude that architecture optimization enables accurate real-time traffic monitoring, guiding our project toward efficient and scalable safety applications.
Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8	Two-step approach: DetectNet_v2 (ResNet18 + TAO) for rider detection, YOLOv8 for helmet + number plate detection, OCR for plate recognition; Custom dataset of 1,715 images	We conclude that combining multi-model deep learning improves reliability. This supports our project vision to use a multi-model approach for accurate detection, even in complex real-time traffic scenarios.

TITLE	TECHNIQUE USED	CONCLUSION
Automated detection of bicycle helmets using deep learning	Computer vision + state-of-the-art object detection algorithms trained on annotated cyclist datasets, real-world traffic video analysis.	We conclude that helmet detection can scale beyond motorbikes to bicycles, proving our project can extend to multiple road user categories for broader safety enforcement.
Smart Road Surveillance: Helmet Detection and Triple Ride Detection Using CNN and YOLO	YOLOv5 for motorcycle, rider, and plate detection; CNN for helmet/no-helmet classification; OCR (Tesseract) for license plate recognition; real-time triple ride detection	We conclude that integrating multiple detection tasks (helmet, license plate, triple ride) into a unified system enhances road safety enforcement.
An AI multitier system with lightweight classifier for automated helmetless biker detection	Custom CNN model BikeNet-12, datasets with helmet/no-helmet/safe-unsafe helmet classes, YOLOv8 for object detection in varied conditions.	We conclude that lightweight models can deliver high accuracy with less computation, highlighting efficiency as a key factor for our system's real-world deployment.

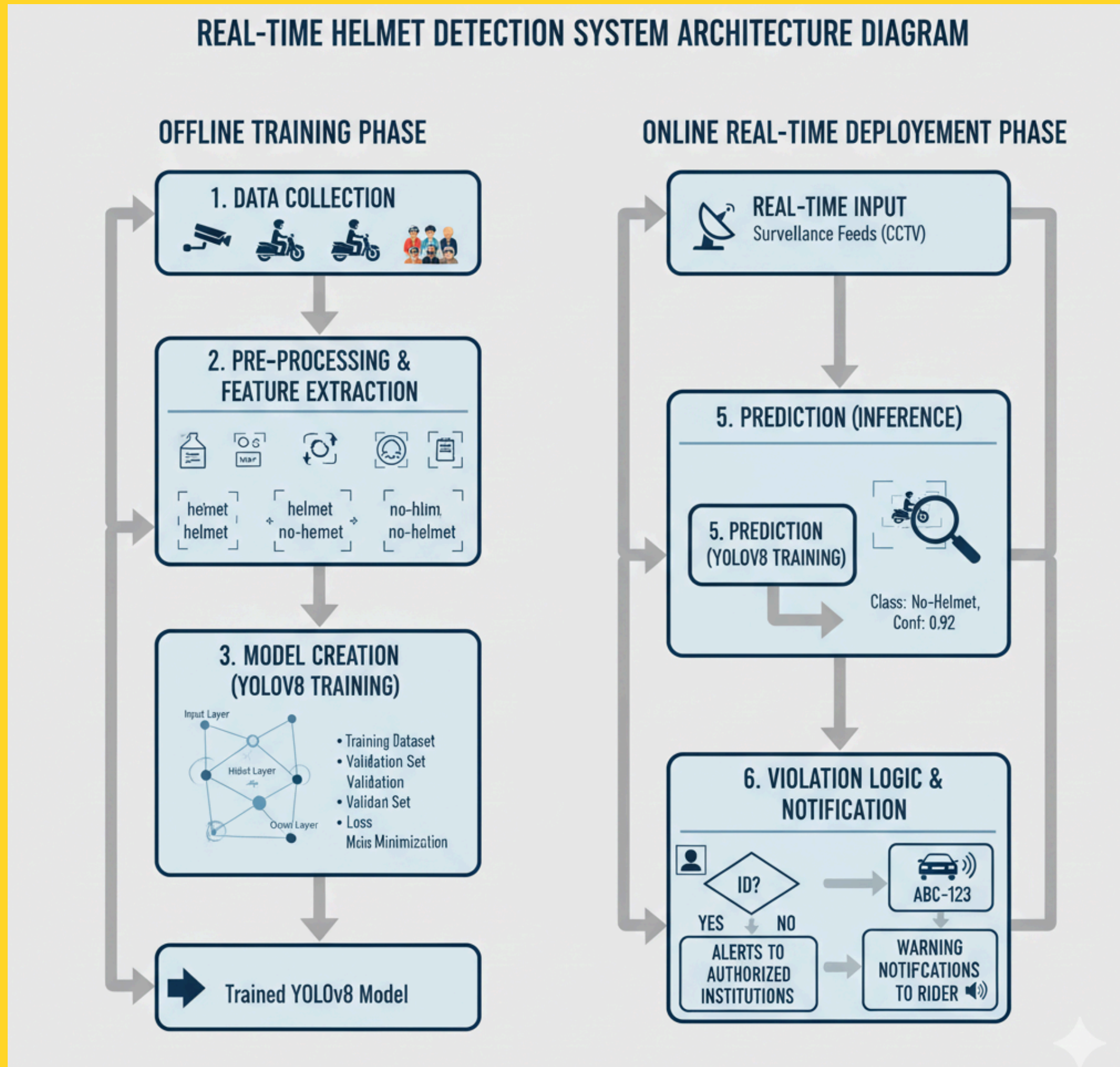
PROPOSED SOLUTION:

- Our proposed work overcomes existing limitations by using YOLOv12 for highly accurate helmet detection across diverse real-world scenarios.
- The system leverages robust datasets and real-time video processing to identify violations instantly. Upon detecting a rider without a helmet, it automatically sends alerts to authorized institutions and issues warning notifications to the rider.
- This ensures reliable, real-time enforcement and supports practical deployment to enhance overall road safety.

PROPOSED SYSTEM:

- The proposed system utilizes the YOLOv12 algorithm to accurately detect helmet usage in real time from surveillance feeds. It is trained on diverse and robust datasets to ensure strong performance across different scenarios.
- When a helmet violation is identified, the system automatically generates alerts and sends them to authorized institutions while issuing warning notifications to the rider.
- This enables precise real-time enforcement and supports practical deployment to enhance overall road safety.

PROPOSED SYSTEM ARCHITECTURE



HARDWARE REQUIREMENTS:

- **Processor : Intel(R) core(TM) i5-10th GEN (or above)**
- **Ram: 4 GB (or above)**
- **Processor speed: 2 GHz**
- **Hard disk: 256 GB SSD (or above)**

SOFTWARE REQUIREMENTS:

- **Operating System: Windows 10 and above**
- **Coding language: Python 3.10.2 and above**
- **Visual studio**
- **Front End: HTML, CSS, JavaScript**
- **Back End: Sql lite**
- **Tensor flow**
- **Keras**

RESULT :

- The YOLOv12 model achieved high accuracy with strong precision, recall, and mAP in detecting helmeted and non-helmeted riders.
- The system demonstrated fast processing speed, enabling efficient real-time helmet detection in traffic scenarios.

DISCUSSION:

- The model demonstrates strong reliability and generalization, performing effectively across diverse real-world traffic conditions.
- Its real-time detection capability makes it practical for deployment in traffic monitoring and enforcement systems.
- Some limitations exist, particularly in low-light conditions and poor image quality, indicating scope for further improvement.

ENVIRONMENTAL SETUP SCREENSHOT:

The screenshot displays a Jupyter Notebook titled 'helmet' in a VS Code editor. The Explorer sidebar on the left shows a project structure for 'HELMET' with folders like 'detected_plates', 'downloads', 'runs', 'test', 'train', 'valid', and 'yolo'. It also lists various image files and a 'data.yaml' file. The main editor area shows the 'helmetcheck.ipynb' file with the following Python code:

```
import sqlite3
import cv2
import numpy as np
import easyocr
import requests
import os
from ipywidgets import FileUpload, Output
from IPython.display import display, HTML, Image
from ultralytics import YOLO

# Initialize YOLO model
model = YOLO("C:/Users/Home/Downloads/helmet/one/runs/detect/train/weights/best.pt")

# SMS API Setup
API_TOKEN = "b47789c5-ebda-435c-a9b9-e4e44d77679c"

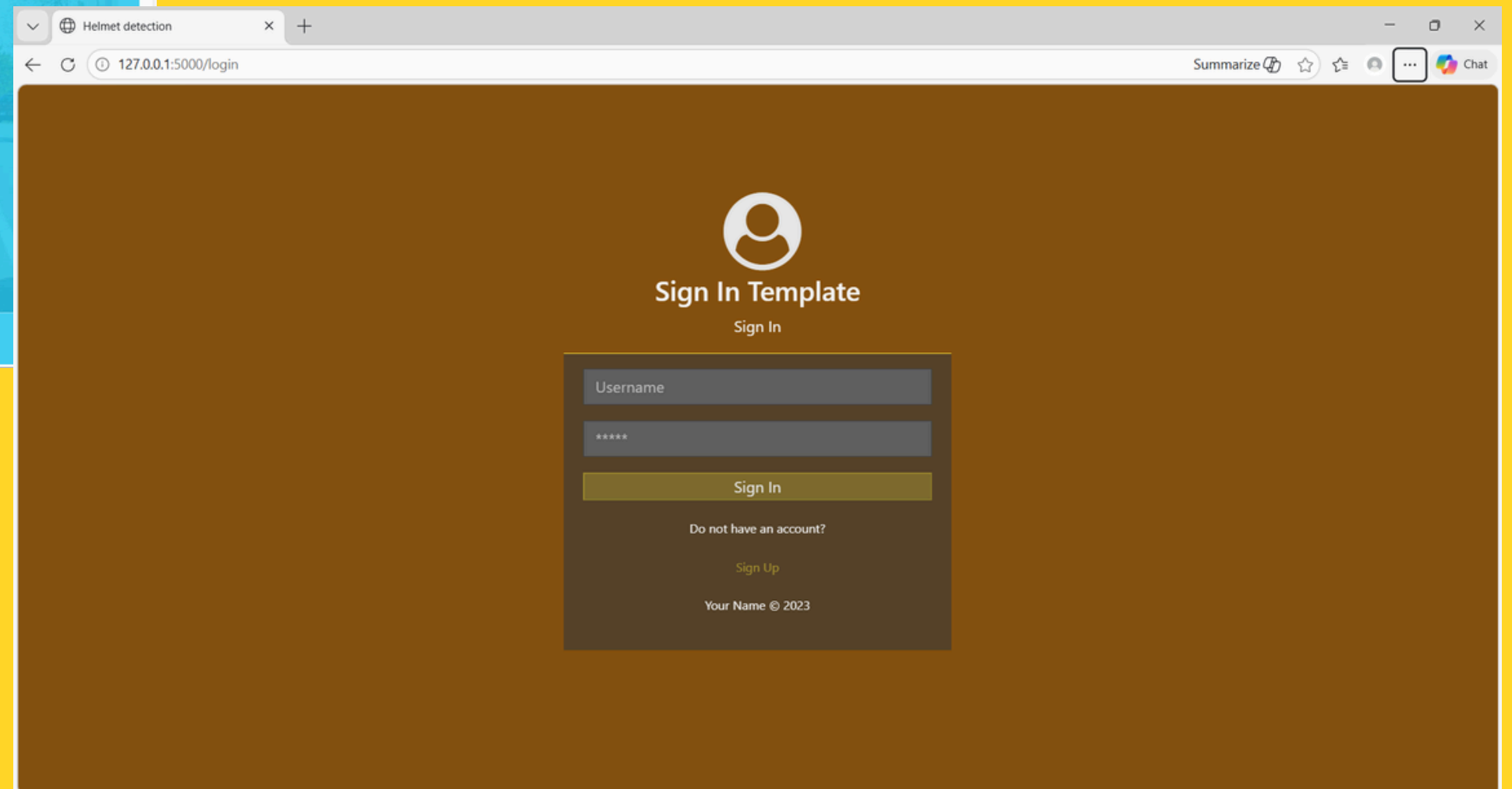
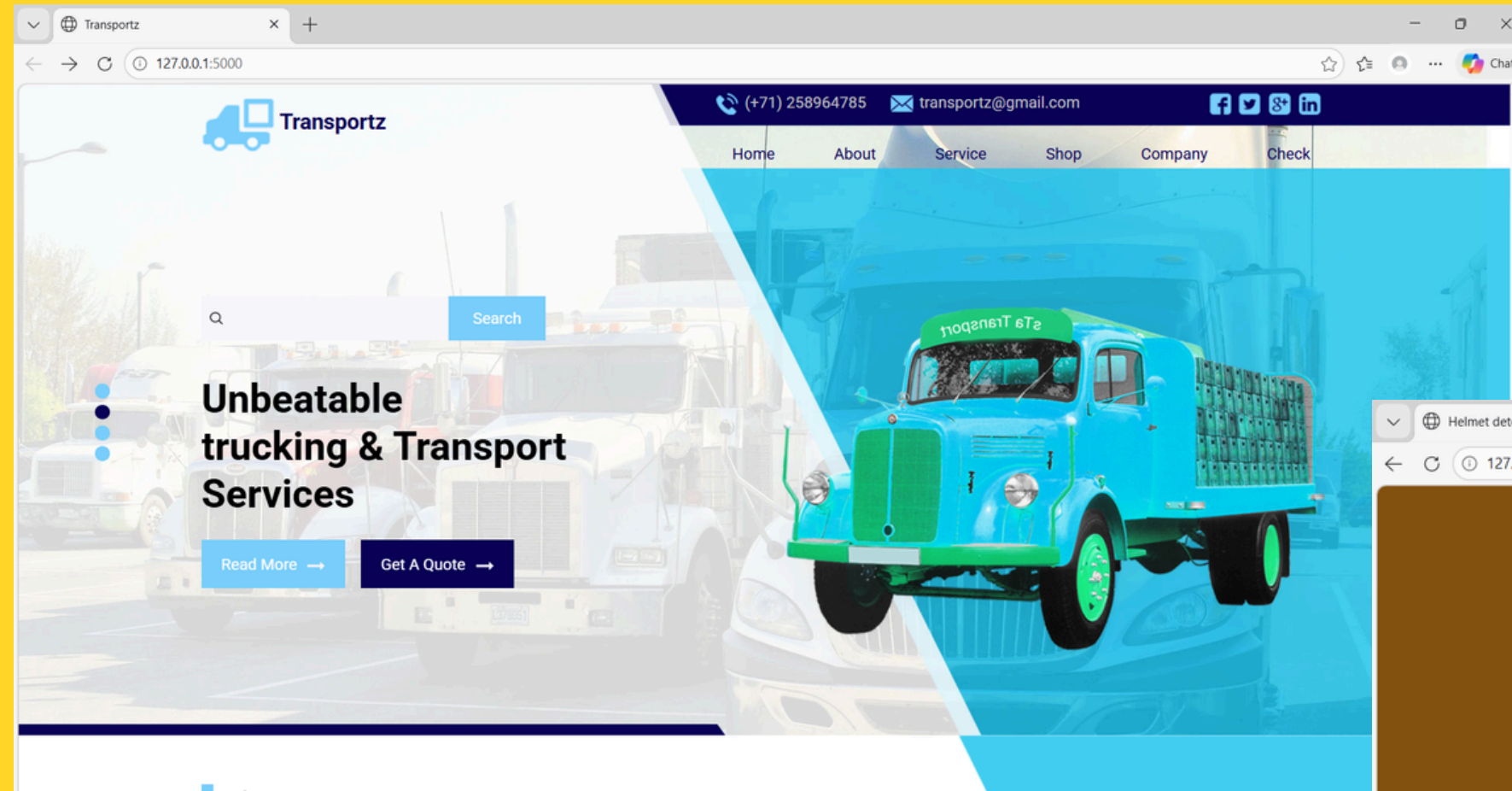
def send_sms(mobile, message):
    """Send an SMS notification."""
    url = "https://whatsbot.tech/api/send_sms"
    params = {
```

The bottom of the notebook shows the 'TERMINAL' output, which includes the following log messages:

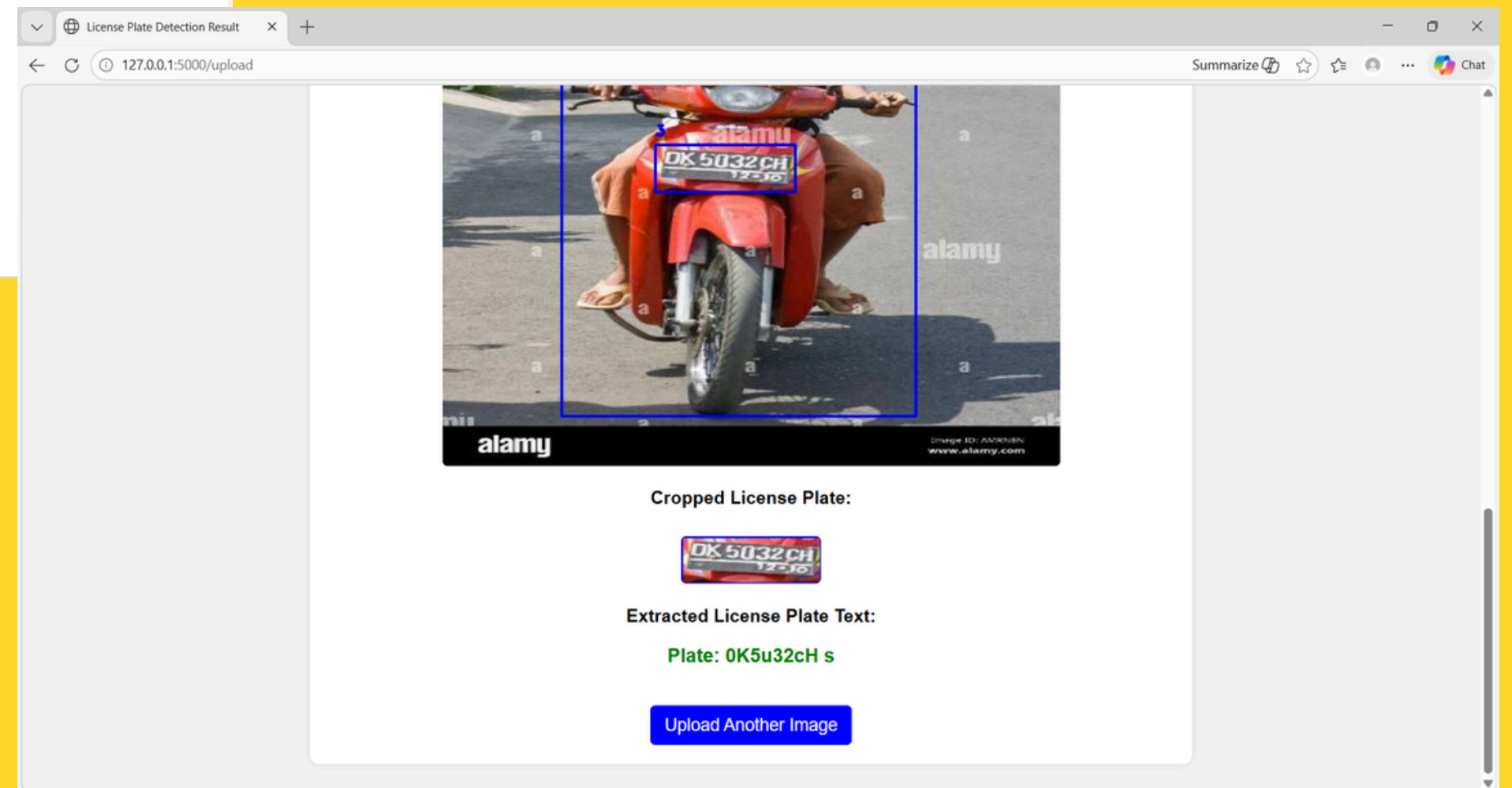
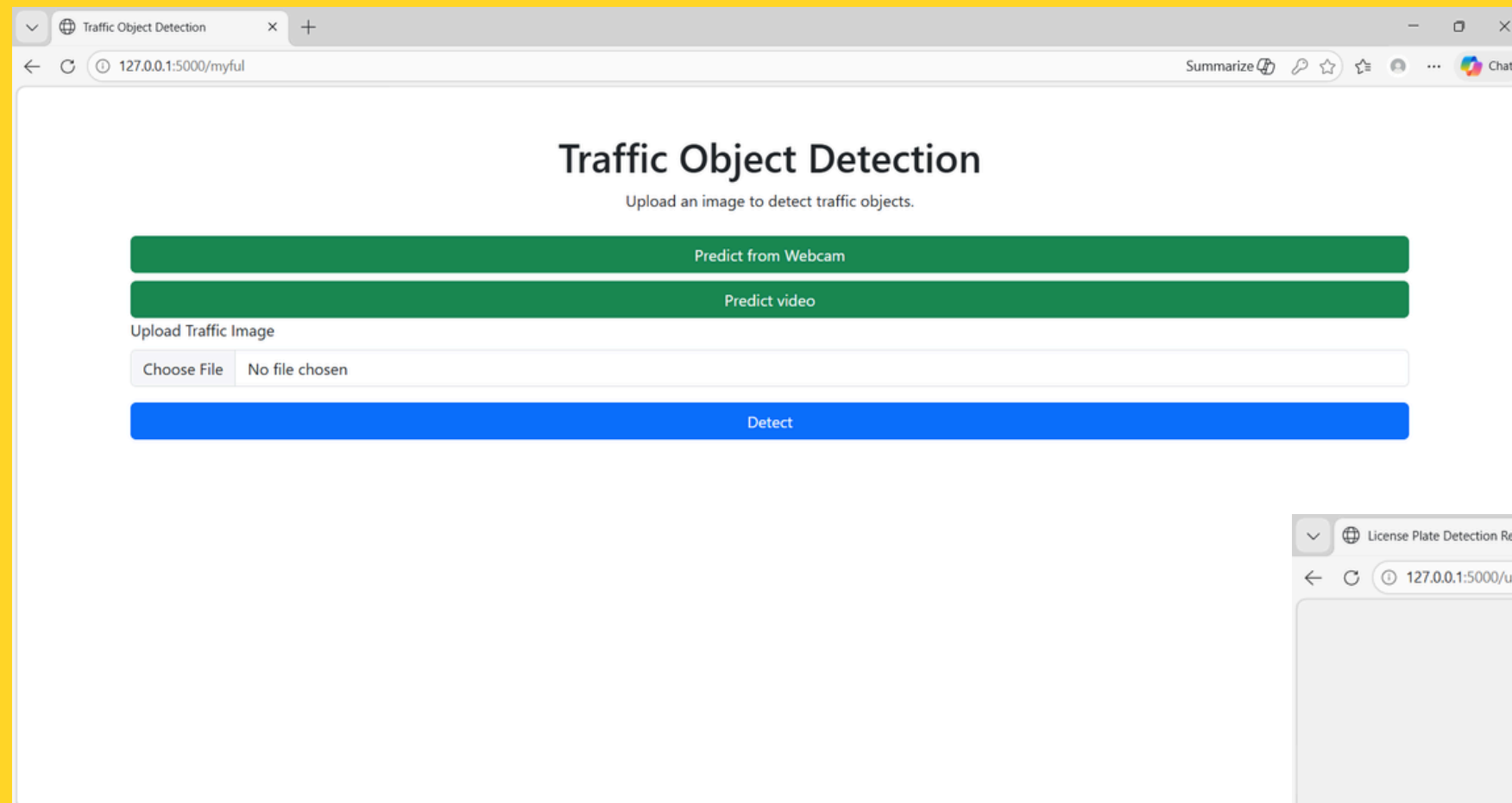
```
0: 640x640 1 1, 2 2s, 1 3, 88.8ms
Speed: 3.5ms preprocess, 88.8ms inference, 1.0ms postprocess per image at shape (1, 3, 640, 640)
Detected class: 2 - 2
Detected class: 2 - 2
Detected class: 1 - 1
Detected class: 3 - 3
License plate detected, processing OCR...
Extracted License Plate: 0K5u32cH s
127.0.0.1 - - [01/Apr/2026 12:05:51] "POST /upload HTTP/1.1" 200 -
127.0.0.1 - - [01/Apr/2026 12:05:51] "GET /static/uploads/4_jpg.rf.ef137f9c8c620e31599f7f3425e70fb4.jpg HTTP/1.1" 200 -
127.0.0.1 - - [01/Apr/2026 12:05:51] "GET /static/uploads/detected_4_jpg.rf.ef137f9c8c620e31599f7f3425e70fb4.jpg HTTP/1.1" 200 -
127.0.0.1 - - [01/Apr/2026 12:05:51] "GET /static/cropped_license_plate.jpg HTTP/1.1" 200 -
PS C:\Users\Siranjeevi\OneDrive\Desktop\helmet\yolo>
```

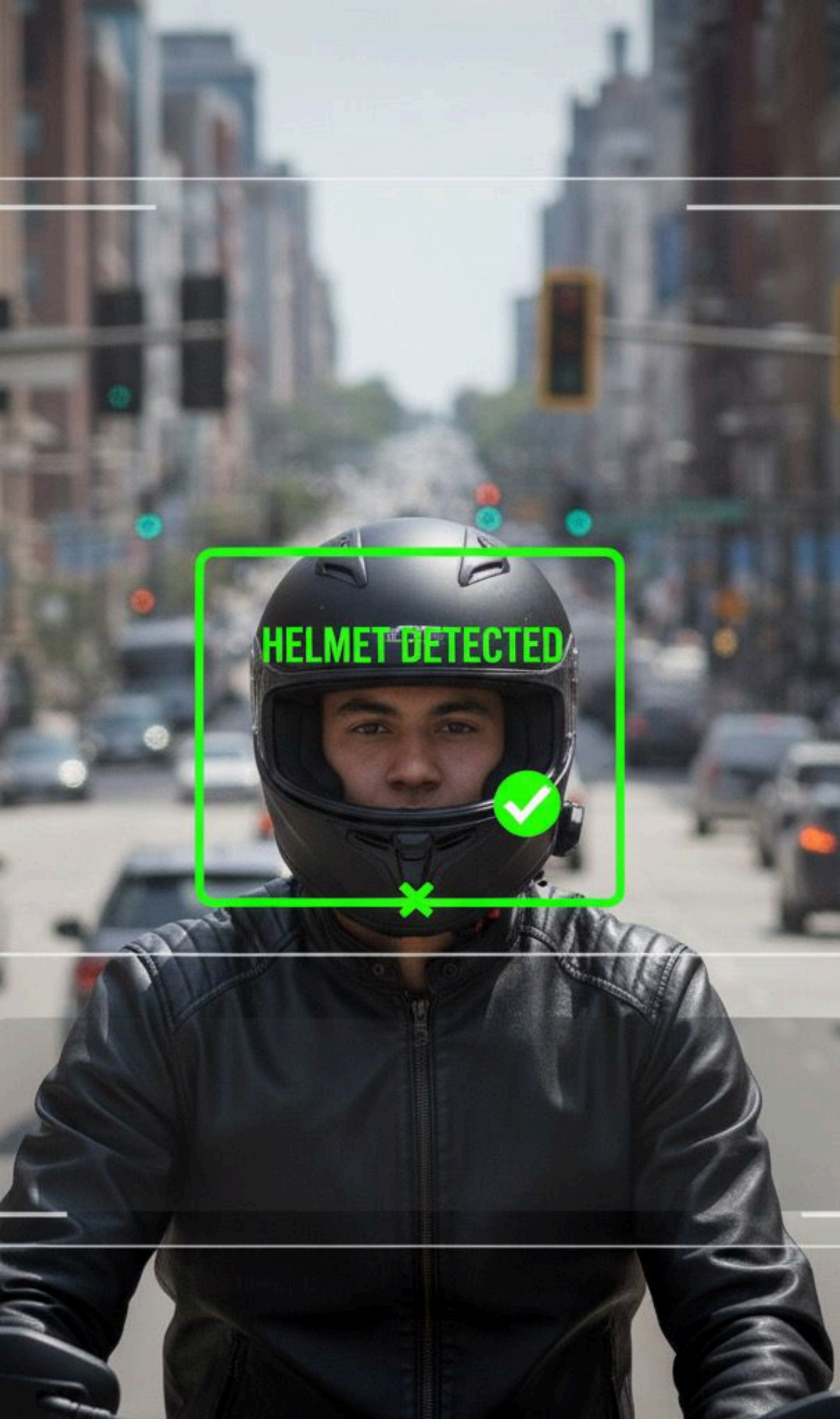
The status bar at the bottom indicates '5 records', 'Spaces: 4', and 'Cell 3 of 3'.

IMPLEMENTATION DETAILS:



IMPLEMENTATION DETAILS:





CONCLUSION:

- We reviewed the helmet detection base paper, explored its advantages, disadvantages, and problem statement, and gained insights into the existing methodologies and challenges involved in real-time helmet detection using deep learning techniques.
- Key limitations in existing systems have been identified, guiding the need for a more practical and efficient approach.
- Core research, planning, and initial development have been successfully completed.
- The Model is under development and accuracy refinement

REFERENCE PAPER:

[01]. Yahia Said, Yahya Alassaf, Refka Ghodhbani, Yazan Ahmad Alsariera.
"AI-Based Helmet Violation Detection for Traffic Management System"

[02]. J. Ranganayaki, Vunnam Poojitha, and Yerroju Dikshith
"Smart Road Surveillance: Helmet Detection and Triple Ride Detection Using CNN and YOLO"

[03]. Uttam U. Deshpande, Goh Kah Ong Michael, Sufola Das Chagas Silva Araujo.
"Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8"

[04]. J.Ranganayaki, Vunnam Poojitha, and Yerroju Dikshith
"Helmet Detection and Triple Ride Detection Using CNN and YOLO"

[05]. Felix Wilhelm Siebert , Christoffer Riis , Kira Hyldekær Janstrup , Hanhe Lin ,Jakob Kristensen"Automated detection of bicycle helmets using deep learning"

CERTIFICATE



Global Conference Hub



of Participation

This Certificate is Presented to

M.SREE HARAN

Computer Science and Engineering, Rajiv Gandhi College of Engineering and Technology.

For attending and giving an oral presentation on the paper entitled

Real Time Helmet Violation and Detection Using Deep Learning

at the **"Third International Conference on Futuristic Trends in Science, Engineering, and Management (ICFTSEM-III) 2026"**- Jointly Organized by School of Management, Xavier University, Patna, India & Global Conference Hub, Coimbatore, Tamil Nadu, India on **27.02.2026 & 28.02.2026**.

August B. Sakay



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August B. Sakay



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The background of the slide features two large, overlapping circles. The circle on the left is yellow, and the circle on the right is orange. The text is centered on the yellow circle.

Thank You for Attending!